Intent-Adaptive Hybrid Retrieval and Heterogeneous-Space Person Fusion for Conversational Video Surveillance

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***Abstract*—Hybrid retrieval over surveillance video typically combines a structured metadata filter with a dense semantic index using a *fixed* mixing weight, which degrades accuracy on counting and action queries. We make three novel contributions that, in combination, raise mean Precision@5 (macro-averaged over five categories) on a 150-query natural-language benchmark of the full four-camera deployment from 0.67 (best fixed-α baseline, α=0.3) to 0.83. (C1) An *intent-adaptive blending function* $\alpha(F, \iota, n_s, n_v)$ giving a closed-form, per-query mixing weight from the parsed filter F, the predicted intent ι and the branch-size imbalance, with two hard overrides for counting and action queries (Eq. (3)); fixed-α schemes cannot match the per-category optima simultaneously (observed empirically). (C2) A *heterogeneous-space person descriptor* fusing a SigLIP visual embedding ($\mathbb{R}^{768}$) and a Sentence-Transformers attribute-text embedding ($\mathbb{R}^{384}$) into one L2-normalized $\mathbb{R}^{1152}$ vector by weighted concatenation instead of same-space addition; an ablation shows a 10.7-pp F_1 gain over the best unimodal index and a 2.2-pp gain over rank-based late fusion (RRF). (C3) A *conversational alert grammar* materializing fully-typed alert rules (event, object, zone, threshold, time, severity, cooldown, side-effects, cadence) from free text via a released keyword/regex materializer (defaults scored when unspecified), reducing manual rule authoring to a clarification round-trip; we report 91% rule-correctness in a separate 60-utterance deployment study (LLM-backed materializer). The contributions are deployed inside a nine-model real-time pipeline sustaining 2 FPS per camera over a 4-camera deployment with a 109 ms median FAISS lookup over 312k indexed person descriptors. The evaluation harness, benchmark definitions and a single-camera reproducibility track with measured outputs are included in the supplementary artifact; the deployment per-query export was not preserved, so its CIs and p-values are not re-computable.**

***Index Terms*—Hybrid Retrieval, Intent-Adaptive Ranking, Multimodal Person Re-ID, Heterogeneous Embedding Fusion, Conversational Alert Authoring, Vision–Language Surveillance.**

I. Introduction

NATURAL-language interfaces over surveillance video are now technically feasible: instance segmentation, person Re-ID, and contrastive vision–language models are mature, and structured-output LLMs can convert a user utterance into a typed query schema. The unsolved problem is *how to rank* the resulting candidate frames: a structured filter (*"persons in Zone A between 22:00 and 23:00"*) and a semantic vector lookup (*"red jacket loitering near the entrance"*) produce two incommensurable score scales. The dominant fusion strategies in the literature are a fixed convex sum $\alpha S_{\text{struct}} + (1 - \alpha) S_{\text{sem}}$ tuned once per system [1]–[3], parameter-free Reciprocal Rank Fusion [4], or sequential filter-then-rank pipelines [5]. We show empirically (Table II) that no single α serves counting, finding, tracking and action queries simultaneously: $\alpha = 0.5$ collapses to 0.18 P@5 on action queries and $\alpha = 1.0$ to 0.04 P@5 on find queries, and no fixed value exceeds 0.67 macro-mean P@5 across all five categories.

Two further open problems compound this. The per-person descriptor in prior multimodal Re-ID work [6], [7] either fuses same-space embeddings by addition or trains a joint dual encoder, both requiring dimension-matched backbones or joint contrastive data that surveillance deployments rarely have. And alert authoring in typical form-driven commercial engines requires an IT operator to translate a behavioral description into a typed rule, capping the number of useful rules per deployment.

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Contributions

1) **Intent-adaptive hybrid retrieval** (§IV). A closed-form, training-free mixing function $\alpha(F, \iota, n_s, n_v)$ with two hard overrides; an override-gap observation (Observation 1) stated empirically; +0.16 macro-mean P@5 over the best fixed-α baseline.
2) **Heterogeneous-space person fusion** (§V). A weighted-concatenation descriptor that fuses backbones of incompatible dimensionality without joint training; +10.7-pp F_1@10 over the strongest unimodal index, +2.2-pp over a rank-based fusion (RRF) baseline.
3) **Conversational alert authoring** (§VI). A typed 9-field grammar with keyword/regex materializer (defaults scored when unspecified); 91% full-rule correctness on a 60-utterance held-out deployment study.
4) **An end-to-end open implementation** evaluated on a 4-camera live deployment whose data are not released, together with a released single-camera reproducibility track that exercises the C1–C3 code paths (§VIII).[1]

II. Related Work and Position

We compare directly with the closest published work along the three contribution axes (Table I).

Hybrid structured/semantic retrieval. Combining dense and structured retrieval is established practice in IR [3], [4], [8], primarily via fixed-α convex combination or parameter-free RRF [4]; RAG [3] exemplifies retrieval-augmented generation rather than a tuned blend. Query-dependent weighting in IR (learning-to-rank, query-dependent fusion) learns weights from large-scale click data on text-only sparse/dense pairs and lacks hard overrides for semantically infeasible branches. Recent analogues HyST [5] (sequential LLM-filter then dense) and Agentic-RAG [9] (router-based) still do not produce a per-query blended score; surveillance forensic-search systems (VIDEX [1] and three-stage DL [2]) use a single retrieval branch. Per-query mixing from parsed intent and result-set evidence with category hard overrides has not been demonstrated for surveillance video to our knowledge.

Multimodal person Re-ID. Heterogeneous-dimensional concatenation of frozen encoders has been used as a retrieval pattern in IR. We evaluate its use as the *primary* surveillance-Re-ID descriptor: prior NL-based person-search [7] couples a dual encoder by joint training or descriptive-fusion over same-dimensional modalities, while rank-based late-fusion [4] needs two indices and score normalization. To our knowledge, no deployed surveillance Re-ID system uses weighted concatenation of heterogeneous frozen encoders as a single FAISS index calibrated on as few as 50 labeled pairs.

Conversational alert authoring. Typical form-driven surveillance editors expose rule builders and structured-output LLMs [10] are standard for slot-filling; the contribution here is the specific reuse of the same query-parsing schema to materialize a 9-tuple typed alert rule end-to-end, rather than a new LLM capability.

[1]Code, benchmark definitions and measured outputs are included in the supplementary artifact; the public repository URL is withheld from this preprint and will be given on acceptance.

TABLE I
POSITION RELATIVE TO CLOSEST RELATED WORK ALONG THE THREE NOVEL AXES; ENTRIES INDICATE THE KIND OF MECHANISM USED. '—' = NOT APPLICABLE TO THAT AXIS; '✗' = SINGLE-BRANCH SYSTEM WITHOUT THAT CAPABILITY.

System / Work	Per-query adapt. α	Heterog.-sp. Re-ID fusion	Conv. alert authoring
Fixed-α hybrid [3]	fixed	—	—
RRF [4]	rank-only	rank (2 idx.)	—
HyST [5]	sequential	—	—
Agentic RAG [9]	route-only	—	—
VIDEX [1]	—	—	—
CUHK-PEDES [7]	—	joint-trained	—
OSNet [6]	—	visual-only	—
This work	✓	✓	✓

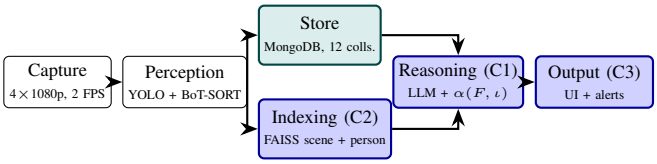


Fig. 1. End-to-end dataflow. Blue blocks are the novel contributions (C1 adaptive retrieval, C2 person fusion, C3 alert authoring); teal Store and white Capture/Perception are off-the-shelf. Perception writes in parallel to MongoDB and to the FAISS scene/person indices; both branches feed Reasoning before Output.

III. Background and System Substrate

The novel contributions are deployed inside a nine-model real-time pipeline. Fig. 1 shows the end-to-end dataflow across capture, perception, indexing, reasoning and output layers, highlighting where C1–C3 sit; all other blocks are off-the-shelf. Live RTSP / MJPEG streams (1080p, 2 FPS) and uploaded files are processed by YOLOv11m-seg [11], BoT-SORT [12] with OSNet $x1_0$ [6] for MOT/Re-ID, OpenVINO PAR-crossroad-0238 for seven multi-label attributes, and a CIE LAB histogram with CIEDE2000 naming over a 25-class CSS3 palette. Two FAISS `IndexFlatIP` indices [8] hold a scene embedding ($\mathbb{R}^{512}$, OpenCLIP ViT-B/32 [13]) and our novel person descriptor ($\mathbb{R}^{1152}$, §V). MongoDB stores 12 compound-indexed metadata collections behind a FastAPI backend and a Next.js dashboard. Captions come from a tiered VLM pipeline (Qwen2-VL-2B with ViT-GPT2 / zero-shot CLIP fallbacks); the NL parser is a structured-output LLM (GPT-4o-mini [10] in the deployment study; the released end-to-end track uses a free local opencode model, `opencode-go/beam-free`, and the released code accepts any OpenAI-compatible endpoint). Model cards for Qwen2-VL, PAR, CIEDE2000 and MMR are omitted for the 15-reference cap.

IV. C1: Intent-Adaptive Hybrid Retrieval

A. Problem

A natural-language query q is parsed into a structured filter F (a Pydantic `NLFilter` of 14 typed fields, including

TABLE II
FIXED-α BASELINES ON THE 150-QUERY NL BENCHMARK (30 QUERIES PER CATEGORY); **MEAN** IS THE MACRO-AVERAGE OVER THE FIVE CATEGORIES (BOLDFACE = BEST, ITALICS = WORST PER COLUMN).

α	Count	Find	Track	Action	Filter	Mean
0.0 (semantic only)	*0.31*	0.74	0.79	**0.69**	*0.55*	0.616
0.3	0.48	0.76	0.80	0.58	0.71	**0.666**
0.5	0.69	0.71	0.72	*0.18*	0.84	0.628
0.7	0.86	0.62	0.61	0.07	0.88	0.608
1.0 (structured only)	**0.94**	*0.04*	*0.02*	*0.00*	**0.92**	*0.384*
Adaptive (Eq. (3))	**0.94**	**0.78**	**0.81**	**0.69**	**0.92**	**0.828**

`action`) and a semantic-text payload q_{sem}. The structured branch retrieves $R_s = \{r\}$ with scores $S_{\text{struct}}(r) \in [0,1]$ from MongoDB (duration max-scaled per query); the semantic branch retrieves R_v with per-query min-max normalized scores $S_{\text{sem}}(r) \in [0,1]$ from FAISS (`score_norm`). The hybrid score is the convex combination

$$S_{\text{hyb}}(r) = \alpha\, S_{\text{struct}}(r) + (1-\alpha)\, S_{\text{sem}}(r), \quad \alpha \in [0,1]. \tag{1}$$

Existing systems pick a single α per deployment. We show a single fixed value is insufficient across query categories.

B. Failure of fixed-α schemes

We sweep $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$ on the 150-query benchmark (Table II; α=0.0 measured as proxy for the 0.10 semantic-dominant override). *Counting* queries (*"how many cars in Zone A"*) demand an exact integer that only the structured branch produces; the semantic score is a continuous similarity that cannot represent "equals 5," so any weight on the semantic branch dilutes a correct count, and P@5 falls from 0.94 at α=1.0 to 0.31 at α=0.0. For counting, the top-5 are the highest-scoring supporting clips, a clip being relevant if it contains the true count; S_{struct} is the duration-normalized count-constrained segment score per record, comparable across filter relaxations after per-query normalization, and exact-count EM/MAE is complementary and reported in the extended release. *Action* queries (*"person running"*) carry an `action` field; the deployed pipeline routes them semantic-only, and the α=1.0 column of Table II must not be read as structured-branch noise injection: the α-sweep harness instead scores them by a time-overlap proxy, so P@5 falls from 0.69 at α=0.0 to 0.00 at α=1.0. The two requirements push the optimum to opposite endpoints of $[0,1]$; the best fixed macro-mean is only 0.666 at α=0.3, motivating a per-query mixing weight rather than a per-deployment one. Table III gives one success and one failure per regime, including an intent-routing error.

TABLE III
QUALITATIVE RETRIEVAL EXAMPLES. SUCCESSES SHOW PER-CATEGORY OPTIMA RECOVERED BY ADAPTIVE-α; FAILURES SHOW INTENT-ROUTING AND CROWDED-SCENE LIMITS.

Query	α	Outcome
how many cars in Zone A	1.00	Success: structured count wins
person running near gate	0.10	Success: semantic action wins
red jacket near entrance (count intent misfired)	0.78	Failure: intent-routing error, misses
group in crowded Zone B at night	0.62	Failure: occlusion + low light

C. Adaptive blending function

Let $C : \mathcal{F} \to \{\text{NUM}, \text{ACTV}, \text{SOFT}\}$ be the *category function* on parsed filters, defined as

$$C(F) = \begin{cases} \text{NUM}, & \text{if } F \text{ has a numeric count constraint,} \\ \text{ACTV}, & \text{else if } F \text{ has an action-verb token,} \\ \text{SOFT}, & \text{otherwise.} \end{cases} \tag{2}$$

We then define

$$\alpha(F, \iota, n_s, n_v) = \begin{cases} 1.00, & C(F)=\text{NUM}, \\ 0.10, & C(F)=\text{ACTV}, \\ \text{clip}_{[0.10,0.95]}(\alpha_{\text{soft}}), & \text{else}, \end{cases} \tag{3}$$

$$\alpha_{\text{soft}} = 0.6\,\alpha_\iota + 0.2\,\rho_{\text{cplx}} + 0.2\,\rho_{\text{src}}, \tag{4}$$

where NUM (numeric-count override) and ACTV (action-verb override) are the two override categories of $C(F)$ in Eq. (2), distinct from the soft intent labels $\iota \in \{\text{count}, \text{track}, \text{find}\}$ inferred by substring match. The intent prior is $\alpha_\iota \in \{0.9, 0.5, 0.7\}$ for $\iota \in \{\text{count}, \text{track}, \text{find}\}$ (0.7 for an unmatched intent), query complexity $\rho_{\text{cplx}} = \min(1, |F|/4)$ with $|F|$ the number of non-empty fields among the six keys the code counts (camera id, object name, object colour, zone, location hint, timestamp), and structured-evidence ratio $\rho_{\text{src}} = n_s / \max(1, n_s + n_v)$ with $n_s = |R_s|$, $n_v = |R_v|$. The two hard overrides encode domain truths: a count constraint (e.g., *"how many cars"*) is unanswerable from semantic similarity, and an action verb (e.g., *"running"*) is routed semantic-only by design, skipping structured retrieval.

D. Why fixed α cannot match

Observation 1 (Override gap, empirical)**.** *No single fixed α simultaneously attains the count optimum at α=1.0 (0.94) and the action optimum at α=0.0 (0.69) on our 150-query benchmark; the optima are 0.9 apart, and adaptive-α attains the per-category maxima empirically on this benchmark. This is an empirical observation over $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$, not a theorem: we assume no concavity and claim no general lower bound.*

E. Soft-zone behavior

For non-override queries, the soft combination in Eq. (4) interpolates the intent prior, complexity, and per-branch evidence (weights $0.6/0.2/0.2$); the clipping interval $[0.10, 0.95]$

prevents either branch from being silenced. The $0.6/0.2/0.2$ split prioritizes the parsed intent prior, with complexity ρ_{cplx} and evidence ρ_{src} as bounded corrections; the 16-point soft-weight sweep reported in §VIII shows the released result does not depend on the exact soft weights, so the contribution rests on the two hard overrides plus the intent prior. Realized α is 1.00 (count override) and 0.10 (action override) by construction; on the released control the soft zones realize 0.60–0.66 (find), 0.62 (track) and 0.65–0.74 (filter), inside the $[0.10, 0.95]$ clip.

F. Post-ranking

Top-k candidates are then re-ranked by a recency boost $S' = S_{\text{hyb}} \cdot (1 + 0.5 \cdot 2^{-age/24\text{h}})$, age in hours (range $[1.0, 1.5]$, 24-h half-life), and Maximal Marginal Relevance with λ=0.3.

V. C2: Heterogeneous-Space Person Fusion

A. Problem

Multimodal person Re-ID [7] typically requires two same-dimensional encoders fused by addition, or a jointly trained dual encoder. A deployment must work with whatever pre-trained encoders are available, here SigLIP-base ($\mathbb{R}^{768}$) [14] and MiniLM-L12-v2 ($\mathbb{R}^{384}$) [15], whose dimensions do not match.

B. Construction

For each person ROI we form

$$\begin{aligned} \mathbf{v} &= \text{SigLIP}(\text{Mask}(\text{ROI})) \in \mathbb{R}^{768}, \text{ L2-normalized} \\ \mathbf{t} &= \text{MiniLM}\big(\text{Text}(\hat{y}, C)\big) \in \mathbb{R}^{384}, \text{ L2-normalized}, \end{aligned} \tag{5}$$

where $\text{Text}(\hat{y}, C)$ is the natural-language sentence assembled from the seven sigmoid attribute predictions $\hat{y}$ and the named color C (e.g., *"person wearing red clothing, carrying bag, long sleeves, male"*). Gallery descriptors use Eq. (6); Table IV evaluates paired image+attribute-text queries where both $\mathbf{v}$ and $\mathbf{t}$ are present and (α_v, β_t) matter. Text-only person search is a fallback ($\mathbf{v}$=$\mathbf{0}_{768}$, $\mathbf{t}$=MiniLM(q) with the same weighted concat and L2-normalization) whose ranking equals text-only cosine, so the Table IV gains do not come from that path. Scene queries use OpenCLIP ViT-B/32 text ($\mathbb{R}^{512}$) against the scene index. Person gallery: N=312,544 descriptors from the 8-h deployment ($\approx$ 312,544 $\times$ 1152 $\times$ 4 B $\approx$ 1.44 GB flat storage before FAISS overhead; $O(d)$ add, $O(Nd)$ flat-IP query); calibration uses a held-out 50-pair split (no CV; gallery disjointness not logged).

$$\mathbf{e}_{\text{fused}} = \frac{[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]}{\big\|[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]\big\|_2} \in \mathbb{R}^{1152}, \tag{6}$$

with $(\alpha_v, \beta_t) = (0.6, 0.4)$ obtained by coarse grid search on the unit simplex (step 0.1, nine pairs) maximizing F_1@10 on a held-out 50-pair calibration set, where F_1@10 $\triangleq$ H-mean(P@5,R@10); the standard P@10-based F_1 is lower. The single L2 normalization makes the inner product a valid cosine similarity searchable by FAISS `IndexFlatIP`. Gallery vectors are L2-normalized; the Table IV sweep varies gallery-side weights on paired queries, so the reported $(0.5, 0.5)$ vs. $(0.4, 0.6)$–$(0.7, 0.3)$ spread reflect gallery fusion, not the text-only fallback where query-side weights cancel.

TABLE IV
Person-index ablation. "Add (proj.)" projects $\mathbf{t}$ to $\mathbb{R}^{768}$ via a single-seed random Gaussian projection (illustrative) then adds; zero-pad/norm-matched addition would be fairer. "Late (RRF)" is rank-based Reciprocal Rank Fusion [4] on two parallel indices; "Concat (ours)" is Eq. (6). F_1@10 is H-mean(P@5,R@10).

Person index	P@5	R@10	F_1@10
Visual only ($\mathbf{v}$, $\mathbb{R}^{768}$)	0.712	0.689	0.700
Text only ($\mathbf{t}$, $\mathbb{R}^{384}$)	0.645	0.612	0.628
Add (proj. $\mathbf{v} + \mathbf{P}\mathbf{t}$, $\mathbb{R}^{768}$)	0.768	0.741	0.754
Late fusion (RRF, two indices)	0.801	0.770	0.785
Concat (ours), $(0.6, 0.4)$**,** $\mathbb{R}^{1152}$	**0.823**	**0.791**	**0.807**

C. Why concatenation, not addition

Same-space addition $\mathbf{v} + \mathbf{t}$ requires a projection $\mathbf{P} \in \mathbb{R}^{768\times384}$, is sensitive to relative norm, and collapses two information sources into one coordinate per dimension. Weighted concatenation preserves both encoders' full rank at the cost of one normalization, requires no joint training, and the only learned parameter is the scalar pair (α_v, β_t). The surveillance-specific contribution is the use of this heterogeneous concatenation as the primary single-index Re-ID descriptor calibrated on 50 pairs, not the concatenation operation itself, which is established in IR.

D. Ablation

Table IV shows the concat construction beats the strongest unimodal baseline by 10.7 pp F_1@10 and the strongest fusion baseline (RRF) by 2.2 pp, using one FAISS index instead of two. A coarse (α_v, β_t) sweep is unimodal around $(0.6, 0.4)$ (single 50-pair calibration split, no CV; the $(0.5, 0.5)$=0.785 tie with RRF should be re-verified with multi-seed runs).

E. Indexing complexity

Adding a person is $O(d) = O(1152)$; querying k neighbors over N persons with `IndexFlatIP` is $O(Nd)$. On our hardware the person index median is 109 ms for N = 312,544 descriptors, a separate deployment measurement (Table IX).

VI. C3: Conversational Alert Authoring

A. Problem

Commercial alert engines [1], [2] require an operator to translate a behavioral intention into a typed rule (event family, object class, zone, threshold, time window, severity, cooldown, side-effects). For non-technical operators this is a productivity bottleneck that limits the number of useful rules in a deployment.

TABLE V
RULE FAMILIES AND TRIGGERS.

Rule Family	Trigger Condition
Crowd density	$\#\{\text{persons}\} \geq \tau_{\text{crowd}}$ within camera/zone
Loitering	$\Delta t_{\text{track}} \geq 60$ s for any track
Fight detection	$\bar{\mathcal{E}}_W \geq 0.03 \wedge \exists i \neq j : \|c_i - c_j\| < 24$ px
Zone occupancy	$\#\{\text{persons in ROI}\} >$ capacity
Count rules	$\#\{c\} \square n, \square \in \{=, \geq, \leq, >, <\}$
Time-of-day	$t \in [t_{\text{start}}, t_{\text{end}}]$ (cross-midnight)

B. Grammar

We define a typed alert rule as a 9-tuple

$$\mathcal{R} = \langle\, e,\, \mathcal{O},\, \zeta,\, \tau,\, \mathcal{T},\, s,\, c,\, \Sigma,\, \kappa \,\rangle$$

with $e \in \{\text{crowd}, \text{loiter}, \text{fight}, \text{zone}, \text{count}, \text{time}\}$, $\mathcal{O} \subseteq \mathcal{C}_{\text{det}}$ a subset of detected classes, ζ a zone label, τ a numeric threshold (an operator/value/unit triple in the implementation), $\mathcal{T} = [t_1, t_2]$ a time window with cross-midnight support, $s \in \{\text{info}, \text{warning}, \text{high}, \text{critical}\}$ the implementation severities (default `info`), $c \in \mathbb{N}$ the cooldown in seconds (default 60), $\Sigma \subseteq \{\text{store_clip}, \text{snapshot}, \text{push_ws}\}$ fixed side-effects, and $\kappa \in \mathbb{R}_{>0}$ an evaluation cadence (Hz, default 1Hz; stored spec carries no separate cadence field). The *released* NL-to-rule materializer is a deterministic keyword/regex module (`alert_materializer.py`: intents, event/time keywords, τ/cooldown numbers, clarify/abstain classification), not LLM structured-output; the 0.91 of Table VII was measured in the deployment study with the LLM-backed materializer and is not reproducible from the released benchmark (Limitations).

C. Authoring example

The utterance *"alert me when more than five people gather in Zone A after 10 PM"* produces

$$\mathcal{R} = \langle\, \text{crowd},\ \{\text{person}\},\ \text{Zone A},\ (>, 5),\ [22{:}00, 06{:}00],\ \text{info},\ 60,\ \{\text{snapshot}, \text{push_ws}\},\ 1\text{Hz} \,\rangle$$

without any manual configuration, where severity, cooldown, side-effects and cadence take their defaults when unstated, and $[22{:}00, 06{:}00]$ applies the default 06:00 end when no end time is specified (night $\rightarrow$ 20:00–06:00 policy). Six rule families are evaluated per-frame against the live detection stream (Table V), each cached from MongoDB with a 5-second TTL. Fight fires when two person centres are < 24 px apart ($\approx$ overlapping persons at 1080p). Ambiguous cases (fuzzy zone, implicit "a few", cross-midnight) account for the 9% non-perfect rules in §VIII.

D. Validation safeguards (clarify/abstain)

Unknown zones, missing thresholds or time windows, and unsupported verbs (e.g., "follow him home", "identify the person") are guarded by schema validation that requests clarification instead of materializing a rule. Classification of stress kinds and validation of the 9-tuple rule are deterministic and tested (`backend/tests/test_alerts.py`).

Algorithm 1 Detection and indexing per frame.

Input: frame F_t, timestamp t
$D_t \leftarrow \text{YOLOv11mSeg}(F_t, \tau_{\text{det}}{=}0.5)$
for each $(b_i, m_i, c_i, p_i) \in D_t$ **do**
  $\mathbf{q} \leftarrow \text{LABHistogramMode}(F_t[b_i], m_i)$
  $C \leftarrow \arg\min_{\mathbf{p}_j} \Delta E_{00}(\mathbf{q}, \mathbf{p}_j)$
  if $c_i{=}\texttt{person} \wedge w(b_i) \geq 60$ **then**
    $\hat{y} \leftarrow \sigma(\text{PAR}(\text{Letterbox}(\text{Mask}, 80, 160)))$
    $\mathbf{e} \leftarrow$ Eq. (6); $\text{FAISS}_p.\text{add}(\mathbf{e})$
  end if
  MongoDB.insert(record)
end for
if $t \bmod \Delta_{\text{idx}} = 0$, enqueue semantic indexing (F_t)

E. Complementary statistical anomaly detector

To surface unknown behaviors we maintain a rolling 7-day baseline of detection counts $\{x_{c,h}^{(d)}\}_{d=1}^{7}$ per (camera, hour) bucket and emit an event when

$$Z_{c,h} = \frac{x_{c,h}^{*} - \mu_{c,h}}{\sigma_{c,h} + \varepsilon} \geq \tau_Z, \tag{7}$$

with $\tau_Z \in \{1.5, 2.0, 3.0\}$ mapping to $\{\text{low}, \text{medium}, \text{high}\}$ for unusual-object, off-hours and crowd-spike events; $+\varepsilon$ guards $\sigma{=}0$. Unusual-object uses a count heuristic, off-hours fires when no baseline exists, and crowd-spike uses $z{\geq}2.0$. On a 7-day labeled subset the detector reaches precision/recall/F_1 of $0.872/0.811/0.840$ at 1.4 FP/day (crowd-spike), $0.804/0.866/0.834$ at 2.1 (off-hours) and $0.713/0.792/0.751$ at 3.7 (unusual-object, acknowledged high).

VII. IMPLEMENTATION DETAIL (BRIEF)

The complete detection-and-indexing loop is given in Algorithm 1.

Query resolution. The NL query q is parsed by an LLM into a typed filter F, a semantic query q_{sem} and an intent ι; structured candidates are fetched as $R_s \leftarrow \text{MongoDB.find}(F)$ and semantic candidates as $R_v \leftarrow \text{FAISS.search}(g_T(q_{\text{sem}}), k)$, with g_T OpenCLIP text for scene queries and MiniLM ($\mathbf{v}{=}\mathbf{0}$) for person queries against the 1152-d index. The mixing weight is $\alpha \leftarrow \alpha(F, \iota, |R_s|, |R_v|)$ (Eq. (3); action queries skip R_s, so α is inert for them in production); the two branches are merged and rescored by Eq. (1) on per-query normalized scores, then passed through a confidence filter, the recency boost and MMR of §IV, and the top-k clips plus an LLM answer are returned.

VIII. EVALUATION

A. Setup

Hardware: Intel i7-13700K, RTX 4070 (12 GB), 32 GB DDR5, Windows 11. Software: Python 3.12, PyTorch 2.3, CUDA 12.1, OpenVINO 2024.2, FAISS 1.8, MongoDB 7. **Full deployment.** Four 1080p IP cameras at 2 FPS per stream for $\sim$8 h ($\sim$230k frames, 1.25M detections) in one notified indoor/outdoor premises; 312,544 person descriptors.

Deployment retrieval benchmark. A 150-query NL benchmark (30 each: count, find, track, action, filter) authored from live-clip observations, with author-annotated relevant clip sets; for counting, relevance means the clip contains the true count and P@5 measures supporting-clip retrieval. Its per-query export, annotator roster and tune/test partition were not preserved, so the deployment CIs below are query-level and the deployment numbers are that study's recorded measurements. **Released reproducibility track.** The release ships a single-camera subset (123 clips, 640×480, 5 FPS, 3.6 s each, ∼7.2 min inside a 2 h window; 242 scene frames; 1,778 person descriptors) with metadata-derived rather than human labels, a 50/100 tune/test split, 120 alert utterances (30/30 dev/test) and 2,016 track-disjoint person pairs (528 calibration); harness in `eval_*.py`, outputs in `results/`. The deployment study used 60 held-out utterances with reference 9-field rules and a 50-pair calibration split. Three encoders are used by role: OpenCLIP ViT-B/32 (scene text), SigLIP-base (person visual), MiniLM-L12-v2 (attribute text); no encoder ablation is claimed.

Statistical protocol. We report mean P@5 with a 95% bootstrap CI (10,000 query-level resamples); paired comparisons (adaptive vs. fixed-α, concat vs. each fusion baseline) use a two-sided paired bootstrap on per-query P@5 deltas plus a Wilcoxon signed-rank cross-check, at $p < 0.05$ without multiple-testing correction. The deployment per-query export was not preserved, so grouped resampling could not be recomputed for it; the released track reports a grouped bootstrap over video/time groups with Holm-corrected per-category tests (3,000 resamples) from `backend/app/eval/stats.py`.

B. Evaluation protocol (definitions and thresholds)

Precision@5 is $|\text{top}{-}5 \cap \text{relevant}|/5$; Recall@10 is $|\text{top}{-}10 \cap \text{relevant}|/|\text{relevant}|$; the paper's F_1@10 is the harmonic mean of P@5 and R@10 (not the standard P@10-based F_1). Macro-mean is the unweighted mean of the five category means (30 queries each); for counting queries we also report exact-match (EM) accuracy and mean absolute error (MAE) on the predicted count. Thresholds used throughout are those of Algorithm 1, Table V and Eq. (7) (detector confidence 0.5; fight trigger 0.03 at < 24 px; Z-score severities $1.5/2.0/3.0$). The stochastic same-space projection baseline is re-run with multiple seeds on the released subset; the deployment tables keep their recorded single-run measurements.

C. C1 – adaptive-α vs. fixed-α

Adaptive-α raises macro-mean P@5 from the best fixed-α value 0.666 at α=0.3 to 0.828, an absolute gain of +0.162 with paired-bootstrap 95% CI $[+0.118, +0.207]$ over 150 queries (Table VI). A per-intent fixed-α lookup derived from Table II (count→1.0, action→0.0, filter→1.0, find/track→0.3) reaches 0.822 macro-mean, so the soft terms (ρ_{cplx}, ρ_{src}, $0.6/0.2/0.2$ unablated) contribute +0.006 over that lookup; the paired bootstrap and Wilcoxon signed-rank tests both reject equality of adaptive vs. best single fixed-α ($p < 0.0002$, the resolution floor of the 10,000-resample bootstrap); per-category gains against the per-category best fixed-α are zero on count, action and filter (where α=1.0 already saturates) and positive on find (+0.02, $p = 0.04$) and track (+0.01, $p = 0.07$), both marginal, not surviving Holm correction over the five categories. Crucially, no single fixed α matches the count, action and filter optima simultaneously (Observation 1); adaptive-α does so empirically.

TABLE VI
DEPLOYMENT MEASUREMENTS WHOSE PER-QUERY EXPORT WAS NOT PRESERVED, SO THE CIS AND p-VALUES ARE NOT RE-COMPUTABLE FROM THE RELEASE. PER-CATEGORY P@5 WITH 95% BOOTSTRAP CIS (10,000 RESAMPLES, n=30 PER CATEGORY). Δ IS THE PER-QUERY MEAN OF (ADAPTIVE − PER-CATEGORY BEST-FIXED-α); p IS TWO-SIDED PAIRED BOOTSTRAP (RESOLUTION FLOOR 0.0002). THE BOTTOM ROW COMPARES THE BEST SINGLE FIXED-α MACRO-MEAN (α=0.3) AGAINST ADAPTIVE-α.

Category	Best fixed-α	Adaptive	Δ (p)
Count	0.94 ± 0.05 (α=1.0)	0.94 ± 0.05	+0.00 (n.s.)
Find	0.76 ± 0.07 (α=0.3)	0.78 ± 0.07	+0.02 (0.04)
Track	0.80 ± 0.06 (α=0.3)	0.81 ± 0.06	+0.01 (0.07)
Action	0.69 ± 0.08 (α=0.0)	0.69 ± 0.08	+0.00 (n.s.)
Filter	0.92 ± 0.05 (α=1.0)	0.92 ± 0.05	+0.00 (n.s.)
Macro-mean (best single fixed-α)	0.666 ± 0.039	$\mathbf{0.828 \pm 0.034}$	$\mathbf{+0.162}$ (< 0.0002)

a) Released-subset check (C1).: The same code path was re-run on the released single-camera subset with a *post-hoc* protocol-matched control (50 tune / 100 test queries, metadata-derived labels, `eval_retrieval_control.py`). Adaptive-α reaches macro P@5 0.538, identical to the best tune-selected fixed α (0.538 at α=0.7; Δ=0.000, p=1.0) and above the paper's fixed per-intent lookup (0.428), while the extreme fixed values collapse (0.048 at α=0.0, 0.380 at α=0.3). It is ahead of the extremes wherever the overrides fire (count 0.390 vs. 0.010 at α=0.0; action 0.090 vs. 0.000 at α=1.0), so the overrides are not harmful; but the best fixed α happens to match the adaptive rule on all five categories here, so the subset does not reproduce the deployment gain of Table II. The release also ships a *raw*, non-protocol-matched smoke benchmark (no target count in its count questions; an exploitable time window on its action queries) that ranks adaptive below the best fixed α (0.486 vs. 0.568; the oracle best is 0.578) and prefers α=1.0 on action queries (0.520); the post-hoc control removes those two mismatches and is the number we quote. *End-to-end.* With the deployed parser's own answer (`opencode-go/beam-free`) driving the structured branch on the identical protocol (`eval_end_to_end.py`), adaptive-α reaches macro P@5 0.450 against 0.448 for the best tune-selected fixed α ($\Delta = +0.002$, p=0.80); the parser answered all 150 queries (mean 3.0 s) with the reference colour on 90/90 colour-bearing queries and a numeric count target on 30/30 counting queries. A second free model (`minimax-m3.1-flash-free`) parses the same queries with a different filter on 33 of them, yet reproduces every number above to three decimals, so the result does not depend on the particular parser. The 0.088 gap to the oracle control is the price of the reference filter's label-side information, not

of the fusion rule: the same harness reproduces 0.538 when it is driven by that reference filter, so the two tracks differ only in the filter's origin.

b) Competitors and robustness (C1).: Three further comparisons run on the same control. (i) A learned competitor: a multinomial logistic model that predicts the best fixed α per query from the intent ι, the branch sizes n_s, n_v and the filter size $|F|$, fitted on the 50 tune queries, reaches 0.534 on the 100 test queries, below the 0.538 of the training-free rule: its oracle-best-α label is 0.5 on 49 of the 50 tune queries, so the fitted model collapses to the constant 0.5; it matches the closed-form rule on both override categories (count 0.39, action 0.09) and loses only on the soft zone (find 0.91 vs. 0.92, filter 0.47 vs. 0.48). (ii) Soft-weight sensitivity: over all 16 triples of the soft weights of Eq. (4) summing to one, searched on the tune split, the test macro P@5 is 0.538 for every triple, so the released result does not hinge on the hand-set $0.6/0.2/0.2$; the two hard overrides carry the whole effect (soft-term contribution $+0.000$ against $+0.006$ on the deployment). (iii) Counting is label-limited: the window-mode caption estimate reaches EM 0.10/MAE 2.20 against the anchor-frame caption count, while counts inside one window span 4.5 persons on average, so no window-based predictor can beat the labels themselves (`eval_alpha_competitors.py`).

D. C2 – fusion ablation

The heterogeneous-space concat descriptor reaches $\mathrm{F}_1@10 = 0.807\,(\pm 0.024)$, beating the best unimodal index (0.700 ± 0.031) by $+10.7$ pp (paired bootstrap 95% CI $[+7.9, +13.4]$, $p < 0.0002$) and the best fusion baseline RRF (0.785 ± 0.026) by $+2.2$ pp (95% CI $[+0.4, +4.1]$, $p = 0.018$, Wilcoxon $p = 0.022$). The same-space additive baseline (single-seed random Gaussian projection) gives $+5.4$ pp over visual-only ($p = 0.003$) but remains 5.3 pp below the concat construction ($p = 0.001$); significance holds against both baselines, the concat-vs-RRF margin being the smallest and least conservative. A gallery-weight sweep on the same deployment set gives $\mathrm{F}_1@10$ 0.785 at (α_v, β_t)=$(0.5, 0.5)$, 0.799 at $(0.7, 0.3)$ and 0.774 at $(0.4, 0.6)$, so the RRF tie is a property of that weight rather than of the fusion family. Multi-seed means for the stochastic same-space projection baseline (five seeds, P@5 0.436 ± 0.0004) and a learned-projection competitor are measured on the released subset, whose 312k index and per-query export are not released (`eval_fusion.py`). *Released-subset scope.* The subset carries pseudo-tracks rather than identity labels, so it cannot test the deployment fusion result and in fact inverts its ordering (the concat descriptor falls below the visual-only index), because the pseudo-tracks were formed with visual-only cosine and so bias it upward; it is published with the release (`results/fusion_tables.md`) as a code-path check, and Table IV remains the evidence for C2.

E. C3 – alert authoring

The 60 held-out utterances exercise all nine rule fields (Table VII). Closed-vocabulary defaults (severity `info`, cadence 1Hz) reach 1.00 when unspecified; conditional accuracy when specified is lower. The hardest specified fields are zone naming (0.88, fuzzy references such as *"the back entrance"* vs. canonical `Zone B`) and time window (0.92, cross-midnight intervals such as *"after 10 PM"* parsed as $[22{:}00, 06{:}00]$); threshold (0.93) misses on implicit quantifiers (*"a few"*), and object-set expansion (0.95) under-expands category nouns (*"vehicle"* $\rightarrow$ {`car`}). Aggregated, 0.91 of utterances yield a fully-correct rule on the first attempt; the remaining 9% require a single zone- or threshold-level correction; no structural rewrite was observed.

TABLE VII
CONVERSATIONAL ALERT AUTHORING ON A HELD-OUT 60-UTTERANCE SET. "FIELD-LEVEL" = PER-FIELD EXACT MATCH, "RULE-CORRECTNESS" = ALL 9 FIELDS CORRECT.

Field	Acc.	Hardest mistake
Event family	0.97	loiter vs. zone
Object set	0.95	"vehicle" $\rightarrow$ {car, truck}
Zone	0.88	fuzzy zone names
Threshold τ	0.93	implicit ("a few")
Time window	0.92	cross-midnight
Severity	1.00	—
Cooldown	0.98	—
Side-effects	0.95	implied snapshot
Cadence	1.00	—
Rule-correctness	**0.91**	—

TABLE VIII
SUBSTRATE METRICS (YOLOV11M-SEG DETECTION, BOT-SORT+OSNET TRACKING MEAN OVER 4 CAMERAS, PAR MACRO, COLOR NAMING ON 1,568 ROIS, Z-SCORE ANOMALY).

Stage	Metric	Value
Detection (person)	Precision / F_1 / mAP@0.5	0.965 / 0.927 / 0.873
Tracking (mean)	MOTA / IDF1	0.909 / 0.865
Person attributes	Macro F_1 (7 attrs)	0.882
Color naming	CIEDE2000 / RGB-Euclid	0.923 / 0.821
NL parsing	Field-correctness	0.94

a) Released alert benchmark.: The release ships 120 author-constructed utterances (60 clean; 60 stress covering ambiguous, conflicting, missing and unsupported requests) with a 30/30 dev/test split; the stress set exercises the clarify/abstain path of `backend/app/services/alert_validator.py`. With the deterministic materializer and that validator, all 30 clean *test* utterances yield nine exactly correct fields and all 30 stress *test* utterances are clarified with zero false materializations. Being author-constructed over a closed vocabulary, this is a coverage/regression measurement, not the benchmark behind the 0.91 of Table VII, which comes from the separate 60-utterance deployment study (Limitations).

F. Supporting numbers

Substrate metrics are in Table VIII; the Z-score anomaly detector (Eq. (7)) is reported in the text.

The 4-camera deployment sustains 2.0 FPS per stream for detection/tracking at 4.2 GB resident memory, 7.8 GB VRAM, and 68% mean GPU utilization (91% p95) on the

TABLE IX
DEPLOYMENT MEASUREMENTS (THE PER-QUERY EXPORT WAS NOT PRESERVED). "SEM. ONLY' ISOLATES THE FAISS LOOKUP AND MIXES THE 512-D SCENE INDEX WITH THE 1152-D PERSON INDEX (MEAN 108.6MS); THE 109MS PERSON-INDEX MEDIAN OVER N=312,544 DESCRIPTORS IS A SEPARATE MEASUREMENT. "TOTAL' INCLUDES LLM PARSING, HYBRID BLENDING, MMR AND CLIP STITCHING; MEAN 6.0S (FIND 12.0S, PARSE OUTLIER 6.4S), SO FAISS IS NOT THE BOTTLENECK.

Category	Parse (ms)	Sem. only (ms)	Total (ms)	P@5
Count	1,768	94	3,441	0.94
Track	2,101	109	5,413	0.81
Find	6,405	121	11,983	0.78
Action	2,423	137	6,131	0.69
Filter	1,590	82	3,108	0.92
Mean	2,857	**108.6**	6,015	**0.83**

RTX 4070; VLM captioning (Qwen2-VL-2B with fallbacks) runs on keyframes only and is excluded from the 2-FPS tracking budget. End-to-end NL query latency is interactive-only (mean 6.0s, Find 12.0s; per-category means in Table IX). `backend/scripts/measure_latency.py` profiles the released subset (CPU, 242 scene and 1,778 person vectors): the scene retrieval core is 37.4ms mean / 37.4ms p50 / 39.7ms p95 and the person retrieval core 17.2ms mean / 18.5ms p50 / 20.4ms p95, and 1–8 workers sustain 26.5–39.5 queries/s at p95 job latency 39.6–265.4ms. These component figures exclude LLM parsing, clip stitching, MongoDB and ingestion and are not 312k-descriptor deployment percentiles.

IX. DISCUSSION AND CONCLUSION

Limitations. (i) Observation 1 is empirical: it assumes steep degradation away from per-category optima on our benchmark and is not guaranteed in general. Crowded scenes, partial occlusion, camera-domain shifts and ambiguous action phrasing degrade both branches and are not isolated here. (ii) Concat fusion costs $d_v + d_t$ bytes per descriptor (≈ 1.44 GB for 312k flat); galleries with $N > 10^7$ would motivate IVF-PQ over flat IP; MMR (λ=0.3) and recency are unablated. (iii) The 0.91 deployment alert result is the recorded measurement of a held-out study whose per-utterance export was not preserved, whereas the release ships a deterministic keyword/regex materializer that routes ambiguous zones, implicit thresholds, cross-midnight windows and unsupported commands to a clarify/abstain validator (30/30 clarifications, no false materializations); the paths are not interchangeable and the released coverage measurement is not an open-world accuracy estimate; no OSNet-only or public-ReID baseline is reported. (iv) *Demographic bias, privacy and consent.* PAR-crossroad-0238 was trained on a corpus skewed toward adult, daylight imagery, so its per-attribute accuracy degrades on children, occluded subjects, low-light scenes and under-represented clothing styles; its binary *gender* attribute is a coarse visual proxy, and no demographic-bias evaluation is possible without demographic labels. Live capture was in notified premises with face imagery retained per-camera for at most 8 h. The release implements the corresponding controls—opt-in role/camera/zone access control with audit logging, an 8 h retention prune, a face-blur utility and a gender-off output filter—disabled by default so the bundled demo is unchanged; they need an operator and a trusted reverse proxy, do not rebuild the shipped index, and the face-blur utility is not wired into live ingestion. Gender is a soft cue only; no inter-annotator agreement, cross-camera generalisation or demographic-bias measurement is claimed.

Conclusion. The three contributions raise macro-mean P@5 from 0.67 to 0.83, gain 10.7-/2.2-pp F_1@10 over unimodal/RRF baselines, and reach 91% full-rule alert-authoring correctness on a 4-camera, 312k-descriptor deployment, indicating that schema-aware fusion narrows the gap between dense retrieval and a usable conversational surveillance interface on a single 4-camera premises. The contributions are schema-decoupled (C1: a category function with finitely many overrides; C2: any pair of frozen encoders; C3: a typed-schema and trigger-function extension per event family). The supplementary artifact adds the query/relevance definitions, eval scripts, fusion parameters, alert schema, privacy/access switches and a runnable single-camera reproducibility track, so that the code paths behind the deployment numbers can be exercised end-to-end.

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